

KEYA LI

Ph.D. Candidate
Transportation Engineering
University of Texas at Austin

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Education

Ph.D.	University of Texas at Austin Transportation Engineering, Department of Civil, Architectural and Environmental Engineering Advisor: Dr. Kara M. Kockelman Dissertation: <i>Mapping Traffic Risk: Driving Behaviors, Exposure, and Safety Interventions</i>	Aug 2027 (expected) GPA: 3.97 / 4.00
M.S.	University of California, Berkeley Transportation Engineering, Department of Civil and Environmental Engineering	Dec 2022 GPA: 3.93 / 4.00
B.E.	Hohai University, China Water and Hydropower Engineering, College of Water and Hydropower Engineering Double Major: Law	Jun 2021 Rank: 2 / 156

Research Interests

Applications: transportation safety and policy, driving behavior, vulnerable road user safety, safety countermeasures, traffic enforcement

Methods: large-scale mobility data analytics, econometrics, spatial analysis, GIS, computer vision

Selected Publications and Manuscripts

Peer-Reviewed Journal Articles

Li, K., & Kockelman, K. M. (2026). Speeding across Texas: Identifying high-risk locations using probe data. *Accident Analysis & Prevention*, 229, 108440. <https://doi.org/10.1016/j.aap.2026.108440>

Under Review

Li, K., & Kockelman, K. M. Exposure-based vulnerable road user safety across U.S. settings: Implications for automated vehicle operations. Under review at *Journal of Safety Research*.

Preprints

Parsa, P., **Li, K.**, Kockelman, K. M., & Choi, S. (2025). Video-based vehicle surveillance in the wild: License plate, make, and model recognition with self-reflective vision-language models. <https://arxiv.org/abs/2508.01387>

Su, G., **Li, K.**, Sengupta, R., & Kockelman, K. M. (2026). Calibration-free vehicle speed estimation: A monocular keypoint-template approach. <https://arxiv.org/abs/2608.16785>

Research and Professional Experience

University of Texas at Austin
Graduate Research Assistant
Austin, TX
Jan. 2024–Present

TxDOT Project 0-7220: Develop Countermeasures to Lower Operating Speeds and Collisions on Arterial Roadways and Reduce Vulnerable User Injuries (with TTI)
Aug. 2024–Present

- Analyzed **39+ million INRIX probe observations** to characterize statewide operating speeds and speeding patterns and examine roadway, land-use, and temporal factors associated with driver behavior.
- Evaluated **non-infrastructure** speed-management countermeasures using before–after analyses of operating speeds and crashes, including Rest-in-Red and roadway lighting as case studies.
- Developed the **Speed and VRU Safety Prediction (SVRP) Tool** to predict operating speeds, speeding rates, and expected VRU crash frequencies and compare countermeasures using estimated benefits, costs, and benefit–cost ratios.

Waymo: Arriving at Pedestrian & Cyclist Crash Rates per Vehicle- and Person-Mile Traveled across US Settings & Modes
Aug. 2024–Dec. 2025

- Developed **pedestrian and cyclist activity models** using **1.1+ million travel-survey trips**, estimating

walking- and biking-miles traveled for **73,000+ U.S. Census tracts**.

- Developed **VRU crash count models for intersections and midblock roadway segments**, integrating exposure, roadway, and land-use characteristics to quantify risk.
- Estimated VRU crash rates in **four U.S. cities with Waymo operations**, providing safety benchmarks for **automated vehicle operations**.

UT Good Systems: Smartphone-Based Method for Speed Limit Enforcement Jan. 2024–Aug. 2024

- Developed and evaluated **smartphone-based computer-vision methods** for speed inference, make and model identification, and license-plate recognition.
- Evaluated speed estimation accuracy using roadside-camera and manually collected smartphone videos.

Center for Transportation Research: Regional Network Modeling Jan. 2024–May 2024

- Built roadway network datasets for Travis, Williamson, and Hays Counties in Texas and refined City of Austin signal timing data for transportation modeling.

Womanium Foundation

Austin, TX

Part-Time Researcher

Sep. 2023–Jan. 2024

- Co-developed a **white paper** for the Jacksonville Transportation Authority on integrating automated vehicles into public transit services.
- Interviewed transportation agencies and industry stakeholders, and provided recommendations on AV deployment, policy, and public-sector implementation.

University of California, Berkeley

Berkeley, CA

Research Assistant

Jan. 2021–Aug. 2023

Equity Implications of Traffic Noise Exposure: A Case Study in San Jose, CA

- Developed **link-level traffic noise exposure models** for San Jose using Mobiliti-simulated traffic flows and speeds across **74,000+ roadway links**.
- Quantified **traffic noise inequities** across communities by separating passenger-vehicle and truck noise and validating modeled levels against field measurements.

Macroscopic Traffic Flow Modeling and Boundary Control

- Implemented **cell-transmission-based traffic flow models** in MATLAB to simulate the spatial and temporal evolution of traffic density across large-scale networks.
- Applied the model to the Oakland road network to evaluate **network boundary control** strategies under various traffic conditions.

Conference Presentations

2026	Li, K., & Kockelman, K. M. “Exploring the Impact of Roadway Lighting on Driver Behavior and Road Safety.” <i>Presentation</i> at Bridging Transportation Researchers Conference (August 2026).
2026	Li, K., & Kockelman, K. M. “Speeding Across Texas: Identifying High-Risk Locations Using Probe Data.” <i>Presentation</i> at Transportation Research Board Annual Meeting (January 2026).
2025 & 2026	Li, K., & Kockelman, K. M. “Methods for Lowering Speeds and Vulnerable Road User Injuries, Without Physical Design Treatments.” <i>Presentation</i> at Bridging Transportation Researchers Conference (August 2025) & <i>Poster</i> at Transportation Research Board Annual Meeting (January 2026).
2025 & 2026	Li, K., & Kockelman, K. M. “Pedestrian Travel Distances and Fatality Rates Across U.S. Settings.” <i>Presentation</i> at Bridging Transportation Researchers Conference (August 2025) & <i>Poster</i> at Transportation Research Board Annual Meeting (January 2026).
2025 & 2026	Parsa, P., Li, K., Kockelman, K. M., & Choi, S. “Video-Based Vehicle Surveillance in the Wild: License Plate, Make, and Model Recognition Using Vision-Language Models.” <i>Presentation</i> at Bridging Transportation Researchers Conference (August 2025) & <i>Poster</i> at Transportation Research Board Annual Meeting (January 2026).
2024 & 2025	Li, K., Malagavalli, J., Goel, L., Wang, T., & Kockelman, K. M. “Smartphone-Based Method for Speed Limit Enforcement.” <i>Presentation</i> at Bridging Transportation Researchers Conference (August 2024) & <i>Poster</i> at Transportation Research Board Annual Meeting (January 2025).

Teaching

Spring 2025	Teaching Assistant, CE 367G: Design and Evaluation of Ground-Based Transportation Systems University of Texas at Austin. <i>Undergraduate level.</i>
Spring 2024	Teaching Assistant, CE 392T: Transportation Economics University of Texas at Austin. <i>Graduate level.</i>
Spring 2022	Teaching Assistant, CE 291D: Data-Driven Control Methods for Civil Engineering University of California, Berkeley. <i>Graduate level.</i>

Mentoring

Summer 2026	Sai Nikhil Pillai , M.S. Student in Information Science, University of Texas at Austin. Mentored research on dashcam-based methods for vehicle speed detection and estimation.
Spring 2025	Gouri Sangal , Undergraduate Student in Civil Engineering, University of Texas at Austin. Mentored research on speed limit verification and literature review of speed-management countermeasures.

Service and Leadership

2026–Present	President , WTS Student Chapter, University of Texas at Austin.
2025–2026	Secretary , Transportation Student Council, University of Texas at Austin.
2024–Present	Reviewer : Transportation Research Board Annual Meeting; journals including <i>Traffic Injury Prevention</i> and <i>Accident Analysis & Prevention</i> .
2018–2021	President , Department Student Union, Hohai University.

Awards and Honors

2026–2027	Talbot S. Huff, Sr. Highway Engineering Graduate Fellowship , Cockrell School of Engineering, University of Texas at Austin.
2026	Outstanding Student Award , Texas District of the Institute of Transportation Engineers (ITE).
2025–2026	Dr. Jentai Yang Fellowship , University of Texas at Austin.
2024–2025	Mary Kate Collins Memorial Endowed Presidential Scholarship , University of Texas at Austin.
2021	Outstanding Undergraduate , Hohai University.
May 2021	Pan Jia Zheng Scholarship , China Society for Hydropower Engineering.
2020	National Scholarship , Ministry of Education of the People's Republic of China.

Technical Skills

Programming: Python, R, SQL, MATLAB, Visual Basic

GIS and Spatial Analysis: ArcGIS Pro, QGIS, GeoPandas

Methods: Statistical modeling, spatial analysis, transportation safety modeling, large-scale mobility data processing

Licensure: Engineer in Training (E.I.T.)

References

Dr. Kara M. Kockelman

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Environmental Engineering

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Dr. Christian Claudel

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